Evaluating Retrieval-Augmented Generation Variants for Biomedical Multiple-Choice Question Answering

Anonymous ACL submission

Abstract

Developing accurate biomedical Question Answering (QA) systems in low-resource languages remains a major challenge, limiting equitable access to reliable medical knowledge. This paper introduces BanglaMedQA and BanglaMMedBench, the first large-scale Bangla biomedical Multiple Choice Question (MCQ) datasets designed to evaluate reasoning and retrieval in medical artificial intelligence (AI). The study applies and benchmarks several Retrieval-Augmented Generation (RAG) strategies, including Traditional, Zero-Shot Fallback, Agentic, Iterative Feedback, and Aggregate RAG, combining textbook-based and web retrieval with generative reasoning to improve factual accuracy. A key contribution lies in integrating a Bangla medical textbook corpus through Optical Character Recognition (OCR) and implementing an Agentic RAG pipeline that dynamically selects between retrieval and reasoning strategies. Experimental results show that the Agentic RAG achieved the highest accuracy (89.54%) with openai/gpt-oss-120b, outperforming other configurations and demonstrating superior rationale quality. These findings highlight the potential of RAG-based methods to enhance the reliability and accessibility of Bangla medical QA, establishing a foundation for future research in multilingual medical artificial intelligence.

1 Introduction

The advancement of Large Language Models (LLMs) such as GPT and LLaMA has transformed Natural Language Processing (NLP) tasks, including Medical Question Answering (QA). These models excel at understanding and generating human-like text, allowing users to ask complex medical questions in everyday language. However, LLMs often produce inaccurate or hallucinated answers, particularly in medical contexts where accuracy is vital (Qin et al., 2025; Qiu et al., 2024).

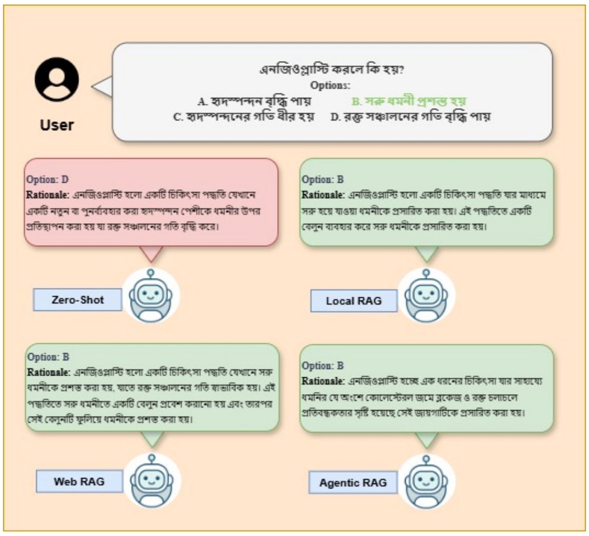

Figure 1: Query responses of llama-3.3-70b-versatile from BanglaMedQA following different strategies.

This limitation arises because these models rely on fixed training data and lack direct access to external, up-to-date knowledge sources.

Retrieval-Augmented Generation (RAG) addresses this limitation by combining information retrieval with generative reasoning to improve answer accuracy (Lewis et al., 2020; Zhang et al., 2024). By fetching relevant documents before generating answers, RAG provides responses that are better grounded in facts, particularly in medical and scientific domains. While RAG has demonstrated strong performance in English biomedical QA systems, its effectiveness in low-resource languages like Bangla, spoken by over 230 million people, remains largely unexplored (Khan et al., 2023; Shafayat et al., 2024; Work, 2023). The lack of high-quality Bangla biomedical datasets and evaluation tools limits the development of reliable QA systems, creating barriers in education and healthcare support (Qin et al., 2025).

Building reliable Bangla medical QA systems involves several challenges. Large-scale Bangla biomedical QA datasets with detailed rationales do not exist, which makes model training difficult (Shafayat et al., 2024; Work, 2023). Translating medical content often distorts meaning and reduces clarity (Pal et al., 2022). Retrieval is also challenging due to sparse Bangla resources (Lewis et al., 2020; Zhang et al., 2024). Finally, evaluating both answer correctness and rationale quality requires specialized metrics and human assessment (Tsatsaronis et al., 2015). Addressing these challenges is essential to expand medical AI to underrepresented languages.

To address these issues, this study develops Bangla biomedical QA datasets and systematically evaluates RAG strategies (see Figure 1). The contributions of this work are as follows:

- We developed BanglaMedQA and BanglaMMedBench, two publicly available Bangla Multiple Choice Question (MCQ) datasets for the medical domain, totaling 2,000 questions.[1] BanglaMedQA contains 1,000 questions from authentic medical admission tests in Bangladesh, each accompanied by reasoning for the correct answer option. BanglaMMedBench includes another 1,000 situational and complex questions, translated and refined from the English MMedBench dataset for Bangla evaluation.

- We implemented and benchmarked multiple RAG variants, including Traditional RAG, Zero-Shot Fallback, Agentic RAG, Iterative Feedback RAG, and Aggregate k-values RAG. We analyzed model performance, rationale quality, and the effects of model size and retrieval strategy in a low-resource Bangla setting.

- For retrieval, we curated an external knowledge base for BanglaMedQA using Optical Character Recognition (OCR) on a standard higher secondary-level Bangla medical textbook used in Bangladesh.

These contributions establish a benchmark for Bangla biomedical QA, evaluate the practical effectiveness of retrieval-augmented strategies, and advance the development of trustworthy medical AI for Bangla-speaking communities. Due to space limitations, related work and detailed experimental settings are provided in Appendix A and Appendix B, respectively.

[1] https://huggingface.co/datasets/ajwad-abrar/BanglaMedQA

2 Proposed Methodology

This section introduces BanglaMedQA and BanglaMMedBench, along with a Bangla biology textbook corpus used as the retrieval source. It then describes multiple Retrieval-Augmented Generation (RAG) strategies, including Traditional RAG, Zero-Shot Fallback, Agentic, Iterative Feedback, and Aggregate Retrieval. Finally, it presents their evaluation on the datasets to assess accuracy, reasoning quality, and the impact of web-based retrieval.

2.1 Datasets

2.1.1 BanglaMedQA: Bangla Medical Corpus

A pioneering contribution, the BanglaMedQA dataset is the first curated collection of Bangla medical question-answering data, sourced from MBBS, BDS, and AFMC admission exams (1990–2024). It comprises 1,000 multiple-choice questions (MCQs), each with four options, correct answers, and detailed rationales, focusing on human physiology and body systems. Quality control measures included:

1. Removing ambiguous or duplicate questions.
2. Excluding non-medical or plant-based items.
3. Ensuring valid option structure and inclusion of rationales.

This dataset serves as a robust, domain-specific benchmark for evaluating Bangla QA and Retrieval-Augmented Generation (RAG) performance in low-resource medical contexts.

2.1.2 BanglaMMedBench: Translated Biomedical Corpus

Building on the MMedBench dataset (Qiu et al., 2024), which includes USMLE-style questions with detailed rationales, BanglaMMedBench was created by batch-translating the dataset into Bangla using the Gemini 1.5B model. The translation process preserved the structural format of the original questions and answers. Post-processing steps ensured translation quality and alignment between answers and rationales.

The final corpus includes 1,000 Bangla biomedical Multiple Choice Questions (MCQs) with rationales more reasoning-intensive than **BanglaMedQA** and serves as a benchmark for evaluating advanced Retrieval-Augmented Generation (RAG) strategies in Bangla.

2.2 Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a framework that combines information retrieval with Large Language Models (LLM). Instead of relying solely on the knowledge stored in a language model's parameters, RAG first retrieves relevant documents or context from an external knowledge source (such as a corpus, database, or textbook) and then conditions a generative model on this retrieved information to produce accurate and context-aware responses. This approach enhances factual accuracy, allows handling of domain-specific queries, and supports low-resource languages by grounding generation in explicit reference material.

2.2.1 RAG Retrieval Source: Bangla Biology Textbook

To support retrieval, the official Bangla biology textbook was digitized and processed. Optical Character Recognition (OCR) was performed using Google Lens, which outperformed Tesseract OCR in handling Bangla conjunct characters and complex typography. Manual verification corrected misalignments, removed formatting inconsistencies, and ensured domain-specific accuracy.

The resulting text corpus serves as a high-quality, machine-readable biomedical knowledge base for Retrieval-Augmented Generation (RAG) evaluation, grounding model responses in authentic Bangla educational material.

2.2.2 Experimental Setup

We evaluated RAG-based approaches for BanglaMedQA and Zero-Shot/Web Search on BanglaMMedBench using LLMs via the Groq API[2] (e.g., llama-3.3-70b-versatile, openai/gpt-oss-120b), with l3cube-pune/bengali-sentence-similarity-sbert for embeddings. The Bangla biology textbook was chunked into 1,000-character segments (200 overlap), indexed with FAISS. Serper API[3] provided web retrieval (top 3 of 8 links). The agentic RAG used a router (Yes/No) with thresholds $\tau 1 = 300$ and $\tau 2 = 200$ characters, falling back to zero-shot if unmet. Iterative evaluation (2 iterations) refined queries, while aggregated RAG (k = 3, 5, 6) used voting (tie-breaker k = 6). Exponential backoff retries (34 attempts) handled API errors.

2.2.3 Implementing Strategies

The methodology shown in Figure 2 integrates local retrieval from a textbook with fallback strategies to external web search or zero-shot reasoning as well as iterative refinement, and aggregation across different retrieval depths to improve answer accuracy and rationale quality.

i) **Traditional/Local RAG:** In this method, a question q with multiple-choice options $O = \{O_A, O_B, O_C, O_D\}$ is input to a Large Language Model (LLM). The model retrieves the top-k passages from the knowledge base D via similarity search:

$$C = \text{Retrieve}(q, D, k)$$

The retrieved context C, question q, and options O form a prompt for the LLM. If C is sufficient (based on a length threshold), the LLM generates a predicted option $\hat{O} \in O$ and rationale R:

$$(\hat{O}, R) = \arg\max_a P(a \mid q, O, C)$$

If C lacks sufficient information, the model outputs:

$$(\hat{O}, R) = (\text{N/A}, \text{উত্তর পাওয়া যায়নি})$$

where উত্তর পাওয়া যায়নি means "The answer was not found" in Bengali. Experiments showed traditional RAG often returns null responses for questions lacking relevant context, especially for chapter-specific queries. In contrast, zero-shot prompting always produces a prediction $\hat{O}$ and rationale R, relying on pretrained knowledge.

ii) **Local RAG with Zero-Shot Fallback:** To address traditional RAG's null responses when retrieved context is insufficient, we developed a hybrid approach combining retrieval with zero-shot fallback, ensuring all questions receive answers.

The pipeline starts with retrieving top-k passages from knowledge base D for query q:

$$C = \text{Retrieve}(q, D, k)$$

[2] Groq API Documentation
[3] Serper API

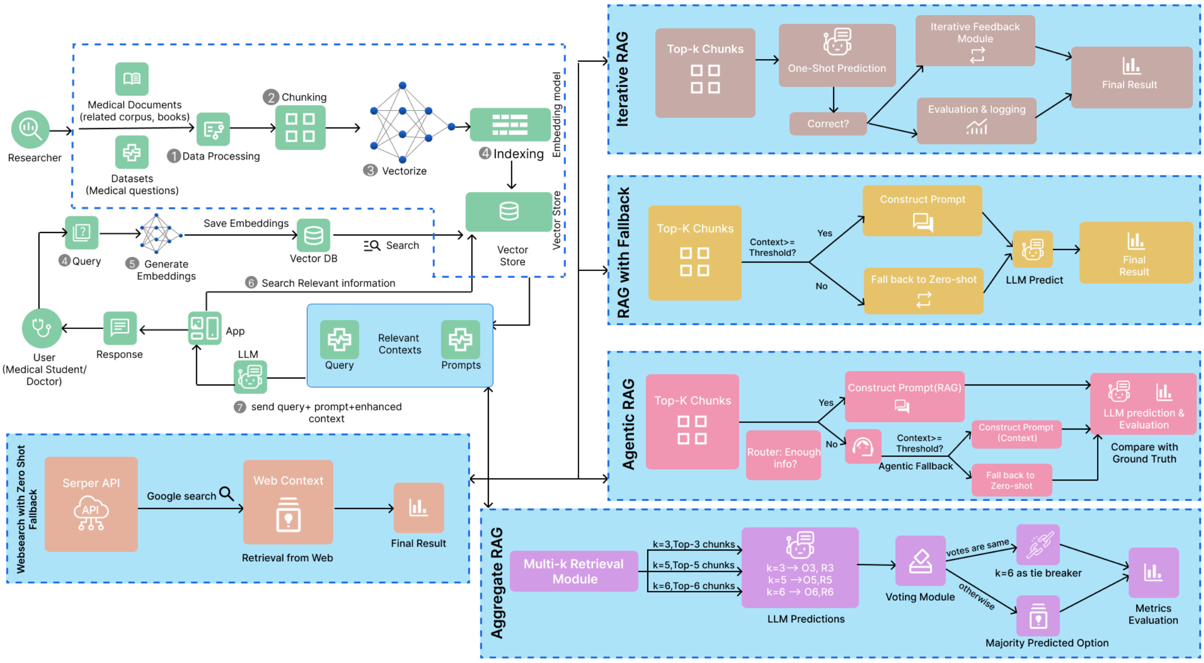

Figure 2: Methodological workflow of the Bangla RAG framework.

The LLM uses q, options O, and context C to predict answer $\hat{O}$ and rationale R:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O, C)$$

If C is insufficient, instead of outputting উত্তর পাওয়া যায়নি ("The answer was not found"), the model falls back to zero-shot, using only q and O:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O)$$

This ensures every question has a prediction, balancing precision (with relevant context) and recall (via zero-shot).

iii) **Web Search with Zero-Shot Fallback:** We developed a Web RAG pipeline for Bangla MCQs, using web retrieval and zero-shot fallback, preserving CSV structure with predictions, rationales, and metrics.

1) Web Retrieval: The question and options are used for a web search via API, retrieving up to eight links. The top three are parsed for text from HTML elements (e.g., articles, paragraphs). Summaries in Bengali bullet points form the web context. If the context meets a length threshold, the model outputs a JSON response:

```
{"O": "A|B|C|D",
 "R": "1-3 line Bangla explanation"}
```

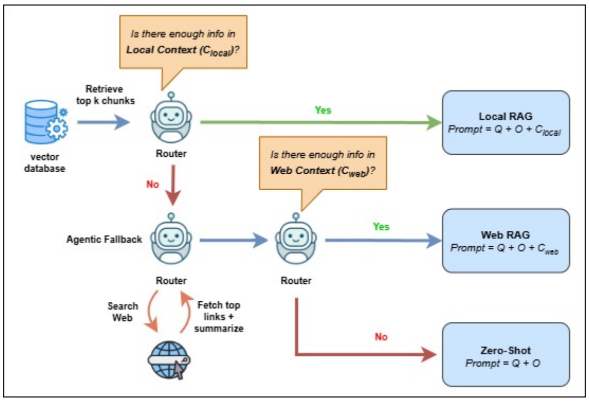

Figure 3: Agentic RAG pipeline illustration.

2) Zero-Shot Fallback: If web context is insufficient, the model uses zero-shot reasoning with pretrained knowledge to generate an answer.

This pipeline leverages dynamic web knowledge for Bangla MCQs, with zero-shot fallback ensuring full coverage and compatibility with evaluation.

iv) **Agentic RAG:** The Agentic RAG pipeline, shown in Figure 3, enhances performance by enabling the LLM to dynamically select the best answering strategy for Bangla MCQs, using local retrieval, web retrieval, or zero-shot fallback.

1) Local Retrieval: The query q retrieves top-

k passages from textbook D:

$$C_{\text{local}} = \text{Retrieve}(q, D, k)$$

A router checks if C_{local} is sufficient using the prompt: "তুমি একটি রাউটার মডেল। নিচের টেক্সটে কি প্রশ্নের উত্তর দেবার জন্য যথেষ্ট তথ্য আছে? শুধু একটি শব্দে উত্তর দাও: হ্যাঁ অথবা না।" , which translates to:

> You are a router model. Does the text below contain enough information to answer the question? Answer in one word only: Yes or No.

If Yes and $|C_{\text{local}}| > \tau_1$, Local RAG generates:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O, C_{\text{local}})$$

2) Web Retrieval: If local context is insufficient, the Serper API fetches up to 8 links, with the top 3 scraped and summarized into Bengali bullet points, forming C_{web}. If $|C_{\text{web}}| > \tau_2$, Web RAG generates:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O, C_{\text{web}})$$

3) Zero-Shot Fallback: If both retrievals fail, the model uses zero-shot reasoning:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O)$$

The routing policy is:

$$\pi(q) = \begin{cases} \text{Local RAG}, & \text{if Router}(C_{\text{local}}) = \text{Yes} \wedge \\ & |C_{\text{local}}| > \tau_1, \\ \text{Web RAG}, & \text{if Router}(C_{\text{local}}) = \text{No} \wedge \\ & |C_{\text{web}}| > \tau_2, \\ \text{Zero-Shot}, & \text{otherwise.} \end{cases}$$

This ensures flexible knowledge selection and full question coverage.

v) **Iterative Feedback RAG:** This method introduces a refinement loop to improve robustness and reasoning within a single query cycle. (see Figure 4)

1) Context Retrieval: The top-k relevant chunks retrieved using similarity search may contain potentially noisy context, which may require refinement in later steps.

$$C_k = \text{Retrieve}(q, D, k)$$

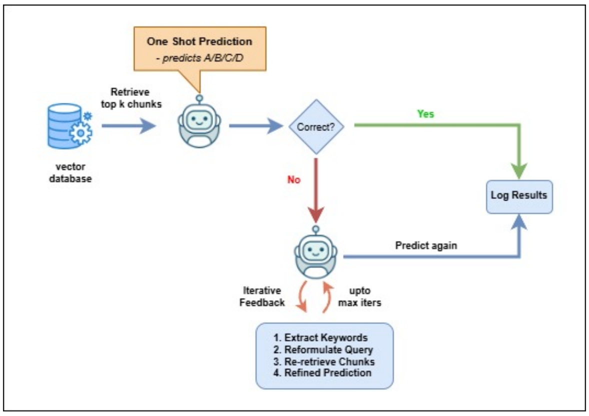

Figure 4: Iterative Feedback RAG pipeline illustration.

2) One-Shot Answering: The LLM generates an initial answer based on the query q and retrieved context C_k:

$$(\hat{O}_0, R_0) = \arg\max P(a \mid q, C_k)$$

If the answer is correct, the process terminates.

3) Iterative Refinement: If incorrect, the LLM extracts key terms, K, from q and C_k to reformulate the query:

$$q' = f(q, K)$$

A new retrieval and prediction are then performed:

$$(\hat{O}_i, R_i) = \arg\max P(a \mid q', C_k)$$

Up to two refinement cycles are allowed to enhance context relevance and correct prior reasoning errors.

4) Final Answer and Confidence: The system assigns confidence levels based on the outcome of the process:

- **High confidence:** Correct one-shot answer.
- **Medium confidence:** Corrected through refinement.
- **Low confidence:** Incorrect even after refinement.

This scoring mechanism enhances interpretability and trust in model outputs.

vi) **Aggregate k-values RAG:** Traditional RAG retrieves a fixed top-k context, making performance sensitive to the chosen k: small k

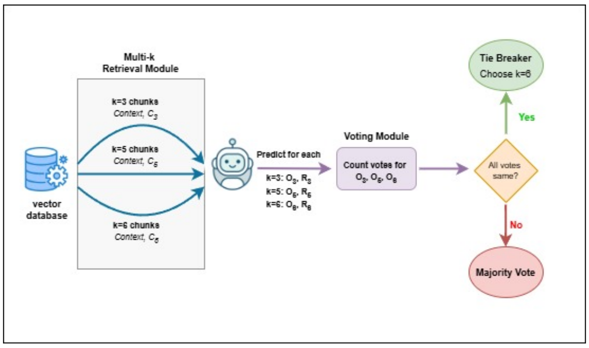

Figure 5: Aggregate k-values RAG pipeline illustration.

may miss key information, while large k can introduce noise. This issue is pronounced in knowledge-dense domains such as zoology. Aggregate k-values RAG mitigates this by combining predictions from multiple k settings to improve robustness and accuracy, as shown in Figure 5

1) Context Retrieval: For a query q over knowledge base D, contexts are retrieved for multiple k values:

$$C_{k_j} = \text{Retrieve}(q, D, k_j), \quad k_j \in \{3, 5, 6\}$$

where different k values balance precision and coverage.

2) LLM Prediction: For each context set C_{k_j}, the model predicts:

$$\hat{O}_{k_j} = \arg\max_{O \in \mathcal{O}} P(O \mid q, C_{k_j})$$

producing an answer option and rationale.

3) Vote Collection and Aggregation: Predicted options $\{\hat{O}_{k_j}\}$ are combined via voting:

- **Majority:** Majority selection is chosen.
- **Unanimous:** Agreement across all k-values indicate high confidence.
- **Tie:** The prediction from $k = 6$ is chosen, leveraging the broadest context.

The final answer is defined as:

$$\hat{O} = \text{Aggregate}\left(\{\hat{O}_{k_j}\}_{j=1}^{m}\right), \quad m = 3$$

The majority rationale corresponding to the winning prediction is selected for consistency and explainability.

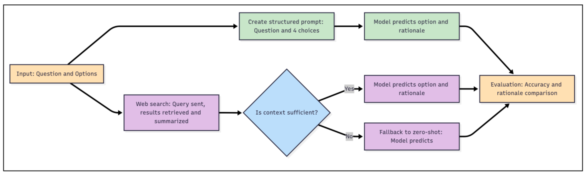

Figure 6: Pipeline for English MMedBench and BanglaMMedBench datasets.

2.3 Evaluating English MMedBench and BanglaMMedBench datasets

We evaluated the MMedBench datasets (English and Bangla) with basic Zero-Shot and Web Search with Fallback methods, as shown in Figure 6, excluding RAG strategies due to its scenario-based, patient-specific questions lacking a reliable textbook corpus. Experiments compared English and Bangla versions to assess translation impact and web search value.

i) **Zero-Shot Analysis:** This method examined model performance on English and BanglaMMedBench datasets, each containing options $O_i = \{O_A, O_B, O_C, O_D\}$, ground-truth answers a_i, and available rationales r_i. For each question q_i, a structured prompt $P_i = f(q_i, O_i)$ was constructed to generate a predicted answer $\hat{a}_i$ and rationale $\hat{r}_i$. Performance was measured using answer accuracy, while generated rationales were qualitatively compared with reference rationales to assess reasoning quality.

ii) **Web Search with Fallback to Zero-Shot:** This pipeline augments web retrieval and zero-shot fallback.

1) Web Retrieval

Query q uses the Serper API for fetching up to 8 links, with the top 3 scraped and summarized into C_{web}. If $|C_{\text{web}}| > \tau_2$, it generates:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O, C_{\text{web}})$$

2) Zero-Shot Fallback:

If $|C_{\text{web}}| \leq \tau_2$ or retrieval fails, it falls back to Zero-Shot:

$$(\hat{O}, R) = \arg\max_{a} P(a \mid q, O)$$

This ensures answers for scenario-based MCQs with limited textbook data.

Model	Accuracy	BERT	METEOR	ROUGE-1	ROUGE-2	ROUGE-L	BLEU-1	BLEU-2
Zero-Shot								
llama-3.3-70b-versatile	73.04%	0.7557	0.1426	0.2063	0.0714	0.1853	0.1414	0.0804
llama-3.1-8b-instant	42.15%	0.7244	0.1149	0.1488	0.0374	0.1299	0.1163	0.0547
openai/gpt-oss-120b	82.49%	0.7300	0.1024	0.1506	0.0325	0.1318	0.1143	0.0509
openai/gpt-oss-20b	**82.60%**	0.7212	0.1048	0.1544	0.0363	0.1362	0.1143	0.0525
Local RAG								
llama-3.3-70b-versatile	82.70%	0.7638	0.1444	0.0603	0.0289	0.0586	0.1294	0.0788
llama-3.1-8b-instant	65.79%	0.7422	0.1451	0.0465	0.0200	0.0457	0.1403	0.0782
openai/gpt-oss-120b	**86.32%**	0.7347	0.1148	0.0825	0.0284	0.0803	0.1215	0.0590
openai/gpt-oss-20b	83.50%	0.7329	0.1220	0.0685	0.0186	0.0675	0.1270	0.0636
Local RAG + Zero-Shot Fallback								
llama-3.3-70b-versatile	82.90%	0.7657	0.1482	0.0560	0.0295	0.0556	0.1366	0.0821
llama-3.1-8b-instant	66.60%	0.7333	0.1517	0.0562	0.0231	0.0541	0.1424	0.0776
openai/gpt-oss-120b	**87.22%**	0.7078	0.1137	0.0796	0.0273	0.0774	0.1234	0.0565
openai/gpt-oss-20b	85.81%	0.6984	0.1193	0.0726	0.0208	0.0700	0.1265	0.0609
Web Search + Zero-Shot Fallback								
llama-3.3-70b-versatile	87.83%	0.7564	0.1469	0.0468	0.0162	0.0458	0.1422	0.0783
llama-3.1-8b-instant	58.35%	0.7141	0.1473	0.0505	0.0222	0.0497	0.1316	0.0679
openai/gpt-oss-120b	**88.83%**	0.7126	0.1048	0.0827	0.0226	0.0805	0.1151	0.0516
openai/gpt-oss-20b	87.02%	0.7122	0.1140	0.0657	0.0199	0.0642	0.1197	0.0571
Agentic RAG (Local + Web) + Zero-Shot Fallback								
llama-3.3-70b-versatile	86.62%	0.7593	0.1476	0.0494	0.0208	0.0483	0.1429	0.0841
llama-3.1-8b-instant	63.38%	0.7238	0.1459	0.0520	0.0177	0.0504	0.1396	0.0749
openai/gpt-oss-120b	**89.54%**	0.7274	0.1136	0.0897	0.0271	0.0870	0.1228	0.0565
openai/gpt-oss-20b	88.83%	0.7353	0.1253	0.0763	0.0228	0.0746	0.1339	0.0669
Iterative Feedback RAG								
llama-3.3-70b-versatile	78.07%	0.7431	0.1424	0.0643	0.0174	0.1462	0.1129	0.0754
llama-3.1-8b-instant	49.20%	0.7982	0.1352	0.0432	0.0343	0.0563	0.1432	0.0723
openai/gpt-oss-120b	**88.73%**	0.7537	0.1348	0.0225	0.0255	0.0801	0.1115	0.0591
openai/gpt-oss-20b	87.42%	0.7231	0.1230	0.0655	0.0176	0.0635	0.1170	0.0646
Aggregate k-values RAG								
llama-3.3-70b-versatile	82.34%	0.6526	0.1211	0.0621	0.0254	0.0532	0.1424	0.0634
llama-3.1-8b-instant	51.81%	0.6321	0.1452	0.2314	0.0732	0.1642	0.1113	0.0697
openai/gpt-oss-120b	**84.51%**	0.6925	0.1059	0.1412	0.0391	0.1227	0.1124	0.0579
openai/gpt-oss-20b	83.95%	0.6688	0.0938	0.1221	0.0310	0.1074	0.0951	0.0453

Table 1: Performance Metrics on BanglaMedQA Across Different RAG Configurations

3 Results and Discussion

3.1 Presenting the Results

Across all methods, as shown in Table 1, Zero-Shot serves as the baseline with accuracies from 42.15% (llama-3.1-8b-instant) to 82.60% (openai/gpt-oss-20b). Traditional RAG improves to 65.79%–86.32%, while RAG with Zero-Shot Fallback reaches 87.22%, and Iterative Feedback RAG offers modest gains but underperforms advanced methods. Agentic RAG achieves the highest accuracy at 89.54% (openai/gpt-oss-120b), showcasing its dynamic retrieval strategy, followed by Aggregate k-values RAG at 84.51% (openai/gpt-oss-120b). Larger models like llama-3.3-70b-versatile outperform smaller variants (e.g., llama-3.1-8b-instant), though gains diminish beyond openai/gpt-oss-20b.

Agentic RAG also excels in rationale quality, with superior ROUGE-1, ROUGE-L(Lin, 2004), and BLEU scores(Papineni et al., 2002), indicating strong linguistic and semantic coherence. Zero-Shot and Iterative Feedback RAG show solid BERTScore(Zhang et al., 2019) and METEOR(Banerjee and Lavie, 2005) values, preserving semantic meaning, while Traditional and Fallback RAG maintain balanced performance. Aggregate k-values RAG remains consistent despite retrieval noise. In summary, Agentic RAG with Zero-Shot Fallback offers the best combination of accuracy and rationale quality, making it the most effective approach for BanglaMedQA.

Table 2 presents accuracy values for the English MMedBench and BanglaMMedBench datasets across models and settings. For Bangla, openai/gpt-oss-120b achieved 90.59%, outperforming llama-3.3-70b-versatile at 62.08%. With Web Search, accuracies dropped to 82.97% and

Model	Bangla Dataset		English Dataset	
	Zero-Shot	Web Search	Zero-Shot	Web Search
llama-3.3-70b-versatile	62.08%	60.00%	88.90%	83.86%
openai/gpt-oss-120b	90.59%	82.97%	92.47%	89.90%

Table 2: Accuracy Values for English MMedBench and BanglaMMedBench Datasets

60.00%, respectively. On the English dataset, openai/gpt-oss-120b reached 92.47% (Zero-Shot) and 89.90% (Web Search), while llama-3.3-70b-versatile scored 88.90% and 83.86%. Zero-Shot performs better on English, while Web Search often reduces accuracy, especially for Bangla, due to scenario-based questions where generic web content offers limited value.

3.2 Interpreting the Results

Traditional RAG improved performance over Zero-Shot by grounding answers in textbook content, confirming that external knowledge enhances factual accuracy. However, gains varied across models due to differences in embedding quality. Llama models showed larger jumps than GPT-oss, which already performed well in Zero-Shot. The fallback strategy brought minor (12%) improvements, mainly by avoiding unanswered questions, though it sometimes retained irrelevant long passages.

Agentic RAG achieved the best overall results, especially with larger models such as `openai/gpt-oss-120b`. Its routing mechanism reduced reliance on context length, enhancing robustness, though occasional errors arose from imperfect routing and noisy web data. Iterative Feedback RAG improved over Zero-Shot but inconsistently; for instance, it reached 78.07% accuracy for `llama-3.3-70b-versatile` versus 88.73% for `openai/gpt-oss-120b`, indicating sensitivity to model capacity and initial retrieval quality. Errors could propagate during iterations, limiting reliability. Aggregate k-values RAG demonstrated stable performance, just below Agentic RAG, by combining predictions across multiple retrieval sizes. Its accuracies 82.34% (`llama-3.3-70b-versatile`) and 84.51% (`openai/gpt-oss-120b`) indicate that aggregating diverse contexts balances precision and coverage, reducing missed information.

On English and Bangla MMedBench, Zero-Shot and Web Search performed better in English, reflecting English-centric model training. In Bangla, `openai/gpt-oss-120b` outperformed `llama-3.3-70b-versatile`, suggesting larger models reduce cross-lingual gaps. However, Web Search sometimes decreased accuracy, as generic web content misaligned with scenario-based clinical questions, indicating that web grounding does not consistently improve performance across domains or languages.

4 Conclusion

This work presents the first Bangla biomedical MCQ system built on two novel datasets, BanglaMedQA and BanglaMMedBench, supported by a Bangla biology textbook corpus. Among various RAG strategies, Agentic RAG with large models such as *openai/gpt-oss-120b* achieved the highest accuracy (about 89%), while Aggregate RAG delivered stable performance (about 84%). Traditional RAG improved over Zero-Shot, with LLaMA models showing larger relative gains. On the translated BanglaMMedBench, *openai/gpt-oss-120b* surpassed *llama-3.3-70b-versatile*, though web retrieval sometimes reduced accuracy due to irrelevant content. Despite limitations in dataset size, translation quality, and computational capacity, the study establishes strong baselines for Bangla biomedical QA and identifies directions for data expansion, better translation, and hybrid retrieval validated by medical experts. Overall, it advances multilingual biomedical reasoning and accessibility for underrepresented languages.

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Appendix

A Related Works

A.1 Medical Question Answering (QA)

Medical Question Answering (QA) systems enable clinicians and patients to access reliable information as medical data grows rapidly. Large Language Models (LLMs) like GPT and LLaMA demonstrate strong performance in Natural Language Processing (NLP) tasks (Qin et al., 2025; Qiu et al., 2024), but biomedical QA remains challenging due to complex terminology and limited domain-specific training. Biomedical QA tasks typically follow two formats:

1. Multiple-Choice Question Answering (MCQ), exemplified by datasets such as *MedMCQA* (Pal et al., 2022) and *PubMedQA* (Jin et al., 2019), which test structured exam-style reasoning.

2. Generative Answering, as seen in benchmarks like *BioASQ* (Tsatsaronis et al., 2015), which evaluate deeper reasoning but are more prone to inaccurate or made-up responses.

These formats highlight the need for robust systems capable of handling diverse medical queries with high accuracy.

A.2 Low-Resource Languages in Medical NLP

While English benefits from extensive biomedical datasets, low-resource languages like Bangla face significant resource scarcity, limiting the development of effective QA systems (Abrar et al., 2024). Efforts such as *BEnQA* (Shafayat et al., 2024), a bilingual science QA benchmark, and Bangla health Named Entity Recognition (NER) (Khan et al., 2023) provide some foundation but lack comprehensive coverage for biomedical QA reasoning. This scarcity underscores the challenge of adapting advanced QA techniques to languages with limited data, particularly in specialized domains like medicine, where accurate and contextually relevant responses are critical.

A.3 Retrieval-Augmented Generation (RAG)

To improve factual accuracy in QA, Retrieval-Augmented Generation (RAG) integrates information retrieval with text generation to ground responses in external knowledge (Lewis et al., 2020). Studies show that retrieval-based models outperform purely parametric ones, enhancing factuality in biomedical QA (Izacard and Grave, 2021; Zhang et al., 2024). However, standard RAG pipelines face issues such as retrieval errors, irrelevant passages, and unstable reasoning. To address these, several RAG variants have been proposed:

1. **RAG with Fallback**: Uses Zero-Shot or parametric generation when retrieval fails (Lewis et al., 2020).

2. **Agentic RAG**: Allows models to decide when to retrieve or reflect, as in *ReAct* (Yao et al., 2022) and *Reflexion* (Shinn et al., 2023).

3. **Iterative Feedback RAG**: Refines retrieval through multiple feedback loops (Shinn et al., 2023).

4. **Aggregate k-Values RAG**: Combines evidence from multiple retrieved passages for robustness (Zhang et al., 2024).

These advancements highlight the potential of RAG to improve QA systems, particularly in challenging domains.

A.4 Research Gap and Motivation

Despite notable progress in Natural Language Processing (NLP) and Large Language Models (LLMs), several gaps remain in the domain of biomedical Question Answering (QA), particularly in multilingual and resource-constrained contexts.

1. **Limited Biomedical QA in Bangla and Low-Resource Languages:** Most existing biomedical QA systems and datasets are developed for English. Low-resource languages like Bangla remain underrepresented, limiting access for a large portion of practitioners, students, and patients in non-English speaking regions.

2. **Scarcity of Domain-Specific Datasets:** Widely used datasets (e.g., LLM-MedQA (Yang et al., 2025), PubMedQA, MedMCQA) are predominantly in English and often based on Western curricula. There is a lack of curated, high-quality biomedical MCQ datasets tailored to South Asian or Bangla medical textbooks, creating a significant barrier for localized applications.

3. **Challenges in Applying General LLMs to Biomedical MCQ Tasks:** Open-domain LLMs (e.g., GPT, LLaMA, Mistral) show strong general reasoning but often lack specialized biomedical knowledge (Jahan et al., 2024). Without domain adaptation, these models struggle with terminologies, abbreviations, and reasoning required for medical MCQs.

4. **Limitations of Current Retrieval-Augmented Generation (RAG) Approaches:** While RAG has improved factual accuracy by grounding LLMs in external knowledge, most biomedical RAG studies focus on research articles and clinical notes, not structured MCQs. Existing RAG pipelines do not adequately handle multilingual retrieval (Bangla–English mix) or exam-style MCQ reasoning with rationales.

5. **Need for Explainability in Biomedical Education:** Current biomedical QA models mostly generate short answers without providing reasoning steps or rationales. For educational and clinical training purposes, explanations are essential, yet underexplored in multilingual biomedical MCQ systems.

This research therefore addresses the above gaps by curating a Bangla biomedical MCQ dataset from textbooks and exam sources, implementing RAG pipelines that handle multilingual retrieval (both Bangla and English) for answer and rationale generation, enhancing both accuracy and explainability.

B Experimental Setup

To evaluate the proposed approaches, we employed multiple Large Language Models (LLMs), external APIs, and supporting libraries to construct a robust RAG-based pipeline. The setup integrates both local retrieval from the Bangla biology textbook and fallback strategies to external web search or zero-shot reasoning.

B.1 Large Language Models

We experimented with a range of models served through the Groq API (Groq, 2025), including llama-3.3-70b-versatile (Dubey et al., 2024), llama-3.1-8b-instant (Vavekanand and Sam, 2024), openai/gpt-oss-20b (Inuwa-Dutse, 2025), and openai-/gpt-oss-120b. The Groq API was chosen for its low-latency inference and ease of switching between different model variants without modifying downstream code.

B.2 Embedding Models

For Bangla text, we used the BengaliSBERT model, a multilingual variant of Sentence-BERT (SBERT) specifically optimized for semantic similarity in Bangla. SBERT modifies the original BERT architecture by introducing a siamese network structure that enables the generation of semantically meaningful sentence embeddings instead of token-level representations. This allows direct comparison of sentence vectors using cosine similarity, which significantly improves efficiency in semantic search and clustering.

B.3 Vector Database and Chunking

The entire Bangla biology textbook was normalized, cleaned, and segmented into overlapping chunks of 1,000 characters with an overlap of 200. This chunking ensured that retrieved passages were semantically complete while avoiding information loss at segment boundaries. A FAISS index (Douze et al., 2025) was built and cached locally for reuse across runs.

B.4 External Web Search

The Serper API (Dev, 2025) was integrated to perform web retrieval. For each query, up to 8 search results were requested, of which the top 3 links were scraped. Extracted raw text was summarized into a compact context before being passed to the LLM.

B.5 Agentic RAG with Router

Unlike traditional RAG, which only checks context length, the agentic pipeline introduced a router module that explicitly judged whether retrieved textbook passages contained sufficient information. The router was prompted to answer Yes or No, and its decision determined whether to proceed with Local RAG or fallback to Web Search. Thresholds were applied to further control routing:

- τ_1 = 300 characters for textbook retrieval.
- τ_2 = 200 characters for web summaries.

If the retrieved or summarized context failed these thresholds, the pipeline defaulted to zero-shot answering.

B.6 Iterative MCQ Evaluation

To improve answer accuracy, questions that failed the initial one-shot evaluation were reprocessed using an iterative method with 2 iterations. In this approach, the model extracted key keywords or short phrases from the retrieved context in each iteration, refining the query before generating the final answer.

B.7 Aggregated RAG Evaluation

To enhance prediction robustness, each question was processed with multiple retrieval settings (k = 3, 5, and 6) from the FAISS vector store. The models answers from these different retrievals were combined using a voting mechanism, with a tie-breaker favoring the largest retrieval set (k = 6). This aggregation approach reduces errors from incomplete context retrieval and improves both option selection and rationale generation.

B.8 Error Handling and Retry Strategy

To handle instability from API calls, the system used exponential backoff retries for Groq and Serper requests (up to 34 attempts). This ensured smoother execution during large-scale batch runs, preventing failures caused by transient network or server issues. In addition, detailed logging was implemented to capture error metadata such as request latency, response status, and endpoint behavior. This information was later used to analyze API reliability trends and optimize retry intervals. Combined with local caching of successful responses, the retry strategy improved both resilience and throughput in the overall evaluation pipeline.

B.9 Evaluation Metrics

To comprehensively evaluate both answer correctness and rationale quality, we utilized a combination of quantitative metrics. Accuracy was used to assess discrete answer prediction performance, while BERTScore, METEOR, ROUGE, and BLEU measured the semantic and lexical similarity between generated and reference rationales.

1. **Accuracy:** Proportion of correctly predicted MCQ answers as in Equation 1, indicating how often the model selects the correct option. Higher accuracy reflects better comprehension and retrieval performance.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{1}$$

2. **BERTScore (F1):** Evaluates the semantic similarity between generated and reference rationales using contextual embeddings from a pretrained BERT model. Instead of relying on surface-level n-gram overlap, it compares embeddings token by token, capturing deeper semantic alignment and meaning preservation between two texts (Zhang et al., 2019). The BERTScore (F1) formula is the standard harmonic mean for F1 scores, calculated as:

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{2}$$

where P is the BERT-Precision and R is the BERT-Recall.

3. **METEOR Score:** Considers precision, recall, and synonym matching for textual similarity. Unlike BLEU, METEOR accounts for word stemming and paraphrasing, making it more sensitive to variations in word choice and sentence structure that still convey equivalent meanings (Banerjee and Lavie, 2005). The METEOR score is calculated as:

$$F_{mean} = \frac{10PR}{R + 9P} \tag{3}$$

$$Penalty = 0.5 \times \left(\frac{\#chunks}{\#unigrams\_matched} \right)^3 \tag{4}$$

$$Score = F_{mean} \times (1 - Penalty) \tag{5}$$

4. **ROUGE Score:** Measures n-gram overlap and longest common subsequence to assess content coverage and informativeness. ROUGE-1 and ROUGE-2 evaluate unigram and bigram overlaps respectively, while ROUGE-L captures the longest sequence of words in common, reflecting overall content similarity and coherence (Lin, 2004).

ROUGE-1 and ROUGE-2:

$$\text{Precision} = \frac{\text{Number of matching n-grams}}{\text{Number of n-grams in candidate summary}} \tag{6}$$

$$\text{Recall} = \frac{\text{Number of matching n-grams}}{\text{Number of n-grams in reference summary}} \tag{7}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

where n-grams refers to unigrams (ROUGE-1) or bigrams (ROUGE-2).

ROUGE-L:

$$\text{Precision} = \frac{\text{Length of LCS}}{\text{Total number of words in candidate summary}} \tag{9}$$

$$\text{Recall} = \frac{\text{Length of LCS}}{\text{Total number of words in reference summary}} \tag{10}$$

$$F_\beta = (1+\beta^2) \times \frac{\text{Precision} \times \text{Recall}}{(\beta^2 \times \text{Precision}) + \text{Recall}} \tag{11}$$

where, LCS is the Longest Common Subsequence and β is a parameter that balances precision and recall.

5. **BLEU Score:** Computes n-gram precision to evaluate fluency and surface-level correspondence with reference text. It rewards outputs that share similar word sequences with reference answers, serving as a traditional benchmark for evaluating the quality and grammatical correctness of generated text (Papineni et al., 2002). The BLEU score is computed using the geometric mean of modified n-gram precisions and a brevity penalty factor.

$$BP = \begin{cases} 1, & \text{if } c > r \\ e^{(1-r/c)}, & \text{if } c \leq r \end{cases} \tag{12}$$

$$BLEU = BP \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{13}$$

where p_n is the modified n-gram precision, w_n are positive weights such that $\sum_{n=1}^{N} w_n = 1$, c is the length of the candidate translation, and r is the reference length.

Rationale for Metric Selection: These metrics were chosen to provide a holistic evaluation of both answer accuracy and rationale quality. Accuracy serves as a direct measure of the models correctness in selecting the right option for MCQs, while BERTScore and METEOR capture semantic and linguistic alignment, assessing whether the models reasoning conveys the intended meaning even if exact words differ. ROUGE metrics evaluate the extent to which the generated rationales cover relevant content from the reference, including structural and sequence similarities. BLEU scores complement this by assessing surface-level fluency and n-gram precision, ensuring that generated explanations are coherent and readable.

By combining these metrics, the evaluation captures multiple dimensions of performance: factual correctness, semantic fidelity, content coverage, and linguistic quality. This is particularly important for assessing models across both English and Bangla datasets, where the goal is to measure not only accuracy but also the quality and clarity of the generated rationales.